

Inverse Functions & Trigonometric Equations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **evaluate** inverse sine, inverse cosine, and inverse tangent expressions.

RELATED SKILLS

☐ I Evaluate an inverse trigonometric expression exactly or approximately.☐ I Recall the principal-value ranges of inverse trigonometric functions.☐ A Express common trigonometric values exactly.

☐ 2. I can **explain** how restricted domains make inverse trigonometric functions possible.

RELATED SKILLS

☐ D Recall the principal-value ranges of inverse trigonometric functions.☐ A Apply the horizontal line test for invertibility.☐ A Exchange domain and range between a function and its inverse.☐ A Test whether a function is one-to-one on a domain.

☐ 3. I can **solve** trigonometric equations exactly or approximately on a specified interval.

RELATED SKILLS

☐ D Isolate a trigonometric expression before solving.☐ D Use reference angles and periodicity to find every solution in an interval.☐ A Check a proposed solution in the original equation.☐ A Determine a reference angle and its quadrant.☐ A Evaluate an inverse trigonometric expression exactly or approximately.

☐ 4. I can **express** all solutions of a trigonometric equation in general form.

RELATED SKILLS

☐ I Express a periodic solution family in general form.☐ D Generate and recognize coterminal angles.☐ A Use reference angles and periodicity to find every solution in an interval.

☐ 5. I can **evaluate** compositions of trigonometric and inverse trigonometric functions.

RELATED SKILLS

☐ D Evaluate an inverse trigonometric expression exactly or approximately.☐ A Recall the principal-value ranges of inverse trigonometric functions.☐ A Substitute one function formula into another.☐ A Verify inverse functions using composition.